MoCHII, M^3, and MAPPINGS V 5.2.1: An Algorithmic and Physical Comparison

Abstract

This memo compares MoCHII (MOnte Carlo for H II regions), the public M^3 (Messenger Monte Carlo MAPPINGS V) tree at `~/RT_Codes/MAPPINGS/M3-main`, and the independent public MAPPINGS V 5.2.1 release at `~/RT_Codes/MAPPINGS/mappings_V-521`. The comparison is based on connected execution paths and quantitative tests rather than on routine names alone. MoCHII is strongest as a modern three-dimensional transport and synthetic-observation engine: octree AMR, an analytic absorption estimator, a dedicated Cartesian DDA backend, solution-driven ionization-front refinement, dust absorption and scattering, stochastic dust/PAH emission, and observer peel-off imaging. MAPPINGS V 5.2.1 is strongest as a deterministic one-dimensional general plasma engine: a 30-element option, density-dependent multilevel cooling, finite-source-age photoionization, non-equilibrium cooling, radiation pressure, and steady radiative MHD shocks with an iterated precursor. Its 1D photoionization path also connects detailed grain/PAH physics to a stochastic infrared solver. M^3 occupies a distinct middle ground: it grafts three-dimensional Monte Carlo transport onto an older MAPPINGS V 5.1.13-era physics tree and therefore combines broad MAPPINGS microphysics with arbitrary Cartesian geometry, but it does not inherit all later 5.2.1 changes or expose all parent modes through its public 3D driver. The principal limitations are correspondingly different. MoCHII presently uses low-density-limit metal cooling fits in its hot thermal loop and has an H II-region-centered ion inventory; M^3 has no AMR and a communication-heavy uniform-grid packet algorithm; and MAPPINGS V 5.2.1 is restricted to spherical or plane-parallel 1D structures, uses an outward-only diffuse-field approximation, and retains a legacy interactive global-state architecture. The recommended strategy is to retain the MoCHII transport architecture, use MAPPINGS V 5.2.1 as the current microphysics and 1D shock/time-dependence reference, and use M^3 specifically to study how MAPPINGS physics behaves after being mapped into three dimensions.

Contents

1. Purpose, scope, and basis of the comparison

MoCHII and M^3 are self-consistent three-dimensional Monte Carlo photoionization codes, whereas MAPPINGS V 5.2.1 is a deterministic one-dimensional plasma and radiative-shock code. MoCHII adds gas physics to the validated MoCafe v2.00 dust radiative-transfer engine. Its founding design treats grid traversal, radiation estimators, dust physics, parallelism, file formats, and synthetic observations as first-class components. M^3 adds three-dimensional Monte Carlo radiation transport to MAPPINGS V, whose principal strength is its extensive atomic, ionic, heating, cooling, and emission-line physics [1, 3]. MAPPINGS V 5.2.1 continues that parent line as version 5.2.1, with separate photoionization, shock, equilibrium-cooling, non-equilibrium-cooling, and single-ion diagnostic drivers.
This memo distinguishes three levels of evidence:

1. *Physics present in the parent code.* The MAPPINGS V source tree contains one-dimensional photoionization, shock, non-equilibrium, dust, and plasma utilities.
2. *Physics connected to the three-dimensional driver.* A routine appearing in the source tree is not automatically active in the `montph7` Monte Carlo path.
3. *Physics quantitatively gated in the current code.* This is the strongest evidence: an analytic test, independent solver, database comparison, or published benchmark with a stated numerical tolerance.

This distinction is essential for shocks and time dependence. The public MAPPINGS tree contains `shock2--shock5`, `timion`, and `timtqui`; however, the three-dimensional `montph7` family sets `nmod='EQUI'` before the cell thermal solve. Thus the present three-dimensional M^3 calculation is an equilibrium photoionization model. The legacy routines remain valuable MAPPINGS capabilities, but they should not be described as coupled three-dimensional M^3 physics without additional development and validation.

No direct same-input, same-atomic-data timing or line-flux campaign among the three local executables has yet been performed. Numerical performance comparisons in this memo therefore use algorithm structure and each code's own recorded gates. They are diagnostic rather than a substitute for a controlled cross-code experiment.

2. Lineage: why MAPPINGS V 5.2.1 is not M^3 5.2.1

The three code names can easily obscure the most important architectural fact. M^3 is not a three-dimensional front end to the newly added MAPPINGS V 5.2.1 directory. Its source identifies itself as MAPPINGS V 5.1.13 with a separate Monte Carlo development history. The two trees still share 78 Fortran filenames, but the shared implementations have diverged substantially; the 5.2.1 versions of core files such as `photo6.f`, `photo7.f`, `shock5.f`, `mapinit.f`, and `cool.f` contain extensive later changes. M^3 adds its own `montph7`, Cartesian packet transport, PDF, and three-dimensional thermal-solver files instead [1, 4].
This gives each branch a different meaning:

- MAPPINGS V 5.2.1 is the newer parent-physics branch. Its public `map52` executable offers P6/P7 multizone photoionization, S5 shock plus precursor models, CIE and NEQ cooling curves, PIE curves, single slabs, and multilevel-ion diagnostic modes.
- M^3 is the three-dimensional geometry branch. It is the relevant comparison for arbitrary density fields, but its physics inventory must be assessed from the connected `montph7` path, not inferred from the presence of a similarly named MAPPINGS routine.
- MoCHII is an independent three-dimensional transport branch built from MoCafe rather than MAPPINGS. Similar physical terms therefore have independent algorithms, atomic-data pipelines, and validation histories.

Consequently, a discrepancy between MoCHII and M^3 cannot safely be labeled a discrepancy with MAPPINGS V 5.2.1 5.2.1. For future differential tests, 5.2.1 should be the primary one-zone and 1D microphysics reference; M^3 should be treated as the older MAPPINGS-derived 3D realization.

3. MAPPINGS V 5.2.1 in its own right

3.1 Deterministic 1D integration and diffuse radiation

P6 and P7 integrate successive zones in spherical or plane-parallel geometry. Step sizes respond to optical depth and predicted changes in temperature and ionization. The result has no Monte Carlo shot noise and is therefore efficient for large parameter grids, detailed spectra, and thin cooling zones. Isobaric models can include radiation pressure, $P(r) = P_0 + P_{\rm rad}(r)$, while other choices provide constant density or a prescribed density law. The model may terminate at a distance, an ion or hydrogen column, or an optical/radiation-bounded condition.
The price is geometric closure. The diffuse field is explicitly labeled "OUTWARD ONLY" in `photo6.f` and `photo7.f`. It follows the allowed 1D directions rather than transporting recombination photons over arbitrary solid angle. This is fast and often adequate for a monotonic sphere or slab, but it cannot represent porous escape, oblique shadows, clump

illumination, or inward/backward diffuse coupling as a true 3D transport calculation can. The word “adaptive” in the S5 menu refers to an adaptive one-dimensional shock integration, not to a 3D AMR mesh.

3.2 Time dependence, cooling flows, and shocks

Unlike the public M^3 3D driver, the P6/P7 path genuinely connects a finite source lifetime to the time-dependent ion solver. It supports an equilibrium solution, a finite-age ionization calculation, and a post-equilibrium source-off evolution. The implementation advances ionic populations with adaptive substeps and couples the elapsed ionization-front time to each new zone. This is a useful one-dimensional ionization-history model, although it is not radiation hydrodynamics and does not evolve an arbitrary multidimensional density field.

The S5 path is an even clearer 5.2.1 advantage. It solves a steady, plane-parallel radiative MHD shock, adapts its step to atomic, cooling, and photon-absorption timescales, and iterates the shock radiation with an automatically calculated photoionized precursor. The source description advertises shock speeds from roughly 10 to 10^4 km s^{-1}. Companion drivers compute CIE cooling, non-equilibrium cooling, and PIE curves. For one-dimensional shock spectra and cooling flows, neither current 3D code offers an equivalent connected and mature path.

3.3 Atomic, electron, and dust physics

The standard preference file activates 16 elements, while `mapFull.prefs` activates all 30 elements from H through Zn; static dimensions permit 31 ion stages and 10,240 photon-energy entries. The thermal loop directly calls fine-structure, intercombination, resonance, multilevel, iron, recombination, free-free, collisional-ionization, charge-exchange, grain/PAH, and cosmic-ray terms. Optional κ electron distributions modify excitation, ionization, and recombination rates. These are major strengths for broad plasma diagnostics and for gas near or above coolant critical densities.

The 1D dust path is materially more complete than the public M^3 3D dust branch. In addition to grain charge, depletion, photoelectric heating, gas-grain collisions, PAHs, and destruction prescriptions, P6/P7 call `dusttemp` zone by zone. That routine implements a Draine–Li stochastic temperature distribution and returns an infrared photon field. Thus MAPPINGS V 5.2.1 has a coupled 1D gas–dust–IR calculation, whereas MoCHII has the stronger spatially resolved 3D dust transport and imaging chain.

Several cautions remain. The atomic tables are broad but globally coupled and are harder to provenance or replace ion by ion than the MoCHII registry. A Compton routine exists but no call from the examined main cooling path was found, and the microturbulent heating block in `cool.f` is commented out; neither should be counted as active 5.2.1 physics without a dedicated execution test. Resonance-line attenuation uses escape/transfer approximations rather than a multidimensional, frequency-redistributing Monte Carlo line solver.

3.4 Software and workflow limitations

MAPPINGS V 5.2.1 preserves a productive but legacy Fortran architecture: common blocks, compile-time maxima, runtime table loading, interactive menus, and many ASCII output products. P7 exposes experimental routines and an API, but the normal workflow remains less scriptable and less modular than MoCHII’s namelist, HDF5/FITS, module, and Python-reader design. The documentation overview describes the manual as work in progress, so reliable use still requires source reading. The benefit of this conservatism is continuity with decades of MAPPINGS models; the cost is a higher barrier to automated regression testing, embedding, and isolated replacement of atomic data.

4. Executive comparison

Table 1: High-level comparison of the current three-dimensional codes.

Topic	MoCHII	M^3
Grid	Octree AMR or a memory-minimal uniform Cartesian backend; optional solution-driven I-front refinement.	Uniform Cartesian grid divided into equal MPI blocks; no AMR.
Absorption transport	Analytic leaf-by-leaf attenuation estimator when the band has no scattering; only direction and frequency are sampled.	Lucy-style random optical depth, discrete absorption, and immediate local re-emission.
Diffuse field	Explicit selected H/He recombination channels rebuilt from the current state; case B OTS remains available.	Local continuum and line emissivity PDF sampled immediately after gas absorption; isotropic re-emission.
Frequency treatment	Dynamically allocated FUV and EUV segments, normally tens of rate-optimized bins with threshold alignment.	Up to 10,240 bins; a fixed number of packets is launched at every frequency and packet energy follows the source spectrum.
Parallel model	Packet decomposition and shared-memory grid arrays; one full grid copy per node.	Spatial domain decomposition; packets crossing blocks are synchronized through MPI.
Gas microphysics	Modern, provenance-bearing fitted data for an H II-region/PDR-centered registry; low-density cooling fits in the thermal loop.	Broad MAPPINGS ion network and multilevel line cooling inside the thermal loop; older but extensively exercised atomic-data compilation.
Dust	EUV/FUV absorption and HG scattering, live dust/PAH survival, equilibrium temperature, SEDust stochastic emission, and IR maps.	Rich grain charge, depletion, photoelectric, and PAH microphysics; the public 3D dust transport branch treats dust extinction/absorption but does not expose an equivalent scattering plus stochastic-IR imaging chain.

Table 1 continued.

Topic	MoCHII	M^3
Synthetic observations	Leaf emissivities, flux-conserving line and field maps, dust-band maps, multi-observer peel-off images, and spectral cubes.	Extensive line spectrum and cell-resolved emissivities; maps can be made by projection, but no equivalent general observer peel-off layer is evident in the public path.
Best current use	Dusty, clumpy, spatially resolved H II regions and atomic PDR skins on simulation-derived density fields.	Broad nebular line diagnostics, denser gas, harder PN-like spectra, and problems that benefit from the full MAPPINGS ion and cooling inventory.

The concise verdict is:

> MoCHII is presently the stronger three-dimensional transport, dust, AMR, and synthetic-observation framework. M^3 is presently the stronger broad-spectrum nebular microphysics framework, especially when many ion stages or density-dependent line cooling determine the solution.

Table 2: What MAPPINGS V 5.2.1 adds to the original two-code comparison.

Topic	MAPPINGS V 5.2.1 strength	Limitation relative to the 3D codes
Photoionization	Noise-free adaptive 1D integration; equilibrium, finite-age, and source-off evolution; radiation-pressure options.	Only spherical or plane-parallel structure; outward-only diffuse field.
Shocks and cooling flows	Connected steady radiative MHD shock plus iterated precursor, CIE cooling, NEQ cooling, and PIE modes.	Steady or prescribed 1D histories, not multidimensional radiation hydrodynamics.
Microphysics	Full 30-element preference set, many ion stages, density-dependent line cooling, κ electrons, cosmic rays, and broad spectral output.	Global tables are harder to audit and update ion by ion; not every dormant routine is connected to the main cooling path.
Dust	Grain/PAH charge and destruction, gas–grain exchange, stochastic temperature distribution, and a coupled 1D IR field.	No arbitrary 3D dust geometry, shadowing, AMR, or observer image construction.

Table 2 continued.

Topic	MAPPINGS V 5.2.1 strength	Limitation relative to the 3D codes
Numerics and workflow	Fast deterministic parameter grids with decades of model heritage.	Legacy interactive common-block architecture, compile-time maxima, fragmented ASCII products, and incomplete manuals.
Best current use	Integrated spectra of 1D H II regions, PNe, AGN slabs, radiative shocks, and equilibrium/non-equilibrium cooling sequences.	Not a replacement for 3D modeling when geometry is part of the physical question.

5. Radiative-transfer algorithms

5.1 Radiation-field estimators

Both three-dimensional codes estimate the cell radiation field from packet trajectories, but their estimators differ fundamentally. M^3 follows the Lucy path length estimator. At frequency ν, a packet with energy ϵ_ν crossing a distance l in cell volume V contributes schematically

$$J_\nu = \frac{1}{4\pi\,\Delta t\,V}\sum_{\text{paths}}\epsilon_\nu l. \tag{1}$$

The distance to the next absorption is sampled from

$$\tau_p = -\ln p, \tag{2}$$

with p uniform on $(0,1)$. The packet crosses Cartesian cells until its accumulated optical depth reaches τ_p, is absorbed, and is re-emitted in situ from the local emissivity distribution. This is conceptually general and treats diffuse radiation naturally.

For the absorption-only ionizing band, MoCHII instead integrates the attenuated packet weight analytically through every crossed leaf:

$$\Delta J_b(\text{leaf}) = wL_{\text{pkt}}\frac{e^{-\tau_{\text{in}}}-e^{-\tau_{\text{out}}}}{\kappa_b}. \tag{3}$$

The packet is followed to the domain edge and contributes the expectation value of the absorbed path in each leaf. There is no sampled absorption location; statistical noise comes only from direction and bin sampling. The analytic attenuation gate measured a mean photo-rate bias of +0.004%. An optional Owen-scrambled Sobol launch (`launch_sequence='sobol'`) stratifies exactly those remaining launch variables: on the same gate the cell-wise photo-rate error then scales as $\sim 1/N$ rather than $1/\sqrt{N}$, with measured effective photon gains of $29\times$ at 2^{17} packets and $222\times$ at 2^{20}, and the launch set becomes independent of the MPI task count. When dust scattering is enabled, MoCHII changes to a forced-first-interaction estimator, applies the dust albedo as a packet weight, samples an HG scattering direction, and uses Russian roulette at low weight.

Equation (3) is a substantial advantage for the present MOCHII target regime: ionizing radiation in H II regions is normally absorption-dominated. M^3 pays the variance of discrete absorption events even in that limit. Conversely, the M^3 packet cycle is more general for a diffuse spectrum containing many continuum and line channels, whereas the current MOCHII diffuse emitter is explicitly limited to selected recombination channels.

MAPPINGS V 5.2.1 has neither Monte Carlo estimator. Its zone integrator is deterministic and therefore essentially noise-free, but achieves that efficiency by imposing spherical or plane-parallel symmetry and an outward-only closure for the diffuse field.

5.2 Diffuse radiation and energy packets

In M^3, gas absorption converts the stellar packet into a diffuse packet immediately. Its new frequency is drawn from a local emissivity PDF containing recombination continua and lines, and its new direction is isotropic. This implements a full three-dimensional alternative to outward-only diffuse-field approximations [1].

MOCHII offers two controlled limits. Case B uses on-the-spot recombination. With the explicit diffuse field enabled, case A is used and the emission table is rebuilt each iteration from the current state. The principal channels are H II, He II, and He III ground recombination continua; the optional He I excited channel adds the 19.82-eV, 584-Å, and two-photon contributions. These packets use the same transport estimator as stellar packets. This split makes case-A/case-B bracketing and energy accounting transparent, but it is not yet the full local emissivity inventory sampled by M^3.

5.3 Spectral discretization

The public M^3 compile-time ceiling is 10,240 frequency entries. The Monte Carlo loop launches N packets for every frequency interval; packet energy is proportional to $L_\nu \Delta\nu / N$. This stratified sampling guarantees representation of weak spectral intervals and is attractive for detailed continua. Its cost scales approximately with $N_\nu N$ rather than with one total packet budget, and its large $J_\nu(x, y, z)$ and opacity arrays are major memory consumers.

MOCHII samples a frequency bin from the luminosity CDF and assigns every packet the same band luminosity. Its bins are designed around rate integrals rather than spectral oversampling. The ionizing grid can align edges with H, He, and active metal thresholds, while an independent FUV segment preserves the EUV resolution. This is efficient for ionization and thermal balance, but a fine line-rich diffuse spectrum would require many more bins and packets than the normal 32–64-bin production setup.

5.4 Velocity fields and line transport

M^3 reads a three-dimensional velocity field and applies a projected Doppler shift to the packet frequency bin during continuum transport. MOCHII currently has no velocity-dependent band transport. This is a real M^3 advantage for bulk-flow effects on continuum opacity.

None of the three codes should yet be treated as a general resonance-line transfer solver. M^3 includes resonance emissivities but, in the published implementation, assigns resonance lines the same cross section as the continuum. This is not a frequency-redistributing resonant-scattering calculation. MOCHII explicitly treats its diagnostic lines as optically thin emissivities and leaves resonance transfer to a dedicated code such as LaRT. MAPPINGS V 5.2.1 applies geometry-dependent escape and attenuation approximations rather than multidimensional redistribution. For forbidden and recombination lines this is normally adequate; for Lyα, C IV, Mg II, or line profiles in moving media, neither current path is sufficient.

6. Grid algorithms, resolution, and parallelism

6.1 Octree AMR versus a uniform Cartesian mesh

The primary spatial difference is adaptive resolution. M^3 accepts an arbitrary density distribution on a Cartesian mesh, but every cell at a given global resolution exists. The published HII20, HII40, and PN150 tests used 33^3 and 55^3 grids. Most integrated lines agreed within 10%, but I-front and boundary tracers such as [O I], [O II], and [S II] could differ by more than 10% [1]. This is the expected weakness of a uniform grid: the thin zone that most needs resolution occupies a small part of the volume.

MoCHII can ingest a pre-existing octree or rebuild the octree after an initial solution. A leaf is refined when it lies in the I-front or sees a large face-neighbor ionization jump. In the recorded gate, refinement from a level-5 start to level 7 produced 464,248 leaves instead of the 2,097,152 cells of a complete level-7 grid; the effective ionized radius agreed with the native level-7 result by 0.008%. The refinement follows clump skins and distributed fronts in turbulent media, not merely the source position.

For problems that truly require a uniform mesh, MoCHII also has a raster-indexed Cartesian backend. It allocates no octree topology, neighbor table, center array, or leaf maps. The dedicated incremental Amanatides–Woo DDA precomputes $t_{\max}$ and $t_{\triangle}$ per ray and advances by comparisons and additions. In the recorded Strömgren test it was twice as fast as the octree traversal, while matching the solution to rounding. At 512^3, omitting the tree and geometry arrays saves approximately 10 GB before physical fields are counted.

6.2 MPI decomposition and scaling trade-offs

M^3 spatially decomposes the Cartesian domain into equal blocks. Each rank stores the local thermal and ionization state, and packets crossing a block boundary are transferred through collective MPI operations. Spatial decomposition is attractive when no node can hold the full grid. It also introduces frequent synchronization: the current transport implementation collectively reduces large packet-state arrays containing flags, positions, directions, frequencies, and absorption counts. At large N_ν, packet count, and MPI task count, communication can dominate.

MoCHII decomposes packets rather than space. The octree and leaf fields reside in MPI-3 shared memory, so ranks on one node share one copy; each node nevertheless holds a complete grid. Ranks transport independent packets and reduce the accumulated radiation tally at the end of a pass. This minimizes packet communication and is well matched to Monte Carlo transport, but the requirement of a full grid on each node ultimately limits the largest possible AMR problem. Thus the scaling comparison is not one-sided: MoCHII should normally scale more cleanly in packet number, whereas M^3 can distribute spatial memory more aggressively. MAPPINGS V 5.2.1 avoids the 3D memory problem entirely by advancing one 1D structure, which is a decisive throughput advantage only when its symmetry assumptions are physically defensible.

7. Ionization, thermal balance, and atomic data

7.1 Element and ion inventories

The local M^3 default atom file activates 16 elements: H, He, C, N, O, Ne, Na, Mg, Al, Si, S, Cl, Ar, Ca, Fe, and Ni. Static array dimensions allow up to 30 elements and 31 ion stages. The published description characterizes the parent MAPPINGS V system as containing 30 elements and roughly 80,000 cooling and recombination lines [1]. The distinction between the parent capacity and the default public data set should be retained when quoting these numbers. The new MAPPINGS V 5.2.1 audit resolves the parent-code side directly: `mapStd.prefs` is the 16-element setup and `mapFull.prefs` activates 30 elements from H through Zn. This full preference set, rather than the M^3 default, is the relevant breadth reference.

MoCHII currently treats H and He plus an eleven-element metal registry: C, N, O, Ne, S, Ar, Mg, Fe, Si, Cl, and Ca. The tracked stages are chosen for H II regions and atomic PDR skins, rather than extending to the highest charge state of every nucleus. New ions are added through provenance-bearing element, cooling-fit, and *n*-level files. This is much easier to audit and extend than the global MAPPINGS tables, but the present inventory is less suitable for hard PN, X-ray, or coronal-ion problems.

7.2 Photoionization, recombination, and charge exchange

MoCHII uses exact hydrogenic ground-state cross sections for H I and He II, and Verner et al. [6] fits for He I and metals. Hydrogen and helium recombination defaults to the Badnell (2023) radiative total minus the Mao–Kaastra (2016) ground-state coefficient (case B), with the Hui–Gnedin fits retained as an option; metal RR and DR are derived primarily from Badnell/CHIANTI v11.0.2 data. Charge exchange uses the modern first-stage rates of Huang et al. and the higher-stage Kingdon–Ferland compilation. Each generated file records its source, fit range, formula, and maximum fit error.

M^3 uses the established MAPPINGS compilation. The published code description identifies photoionization data from Seaton, Raymond, Osterbrock, and Gould, including Auger corrections, and separates hydrogenic from non-hydrogenic recombination. Much of this compilation is older than the MoCHII sources, but it spans more ions and has been exercised in many MAPPINGS applications. “Newer” is not automatically equivalent to “more accurate”: line predictions can differ by several to tens of percent among reputable collision-strength and *A*-value compilations, especially for iron. The decisive test is therefore ion-by-ion comparison with direct database evaluation and observation, not vintage alone.

7.3 Electron distributions and secondary processes

Both M^3 and MAPPINGS V 5.2.1 support either a Maxwellian or a κ electron energy distribution, with rate corrections supplied by KAPPA data. This is useful when exploring non-Maxwellian interpretations of nebular diagnostics. MoCHII assumes a Maxwellian distribution.

All three codes include secondary-ionization concepts. MoCHII optionally partitions the energy of photoelectrons above 40 eV among heat and secondary H/He ionizations following Shull–van Steenberg. The MAPPINGS ion-balance routines call a secondary-ionization module and also include cosmic-ray ionization/heating. The MAPPINGS-derived codes therefore have the broader general-plasma framework. A Compton routine is present in both source trees, but no call from either examined main cooling path was found; it should not be counted as active physics without a dedicated test.

7.4 The critical difference: density-dependent metal cooling

This is the most important gas-physics advantage of both MAPPINGS-derived codes over MoCHII. Their thermal solvers call fine-structure, intercombination, multilevel, and iron line modules during every evaluation of the net cooling function. Level populations and collisional de-excitation therefore affect the equilibrium temperature when n_e approaches or exceeds a transition's critical density.

MoCHII deliberately separates two products:

- analytic Tier-1 cooling fits, evaluated cheaply inside the thermal bisection in the optically thin low-density limit; and
- Tier-2 n-level statistical-equilibrium data, evaluated on the converged state to produce density-dependent diagnostic emissivities.

The output line ratios therefore respond to density, but collisional quenching does not feed back into the hot thermal loop. At ordinary H II-region densities this is often acceptable. In compact H II regions, dense PN, shocked cooling zones, or any model with $n_e \gtrsim n_{crit}$ for a dominant coolant, M^3 is physically better founded. Moving the generic MoCHII n-level solver or a reduced density-suppression fit into the thermal loop is the highest-value gas-physics improvement suggested by this comparison.

7.5 Thermal processes and equilibrium solution

All three codes solve ionization and temperature under the current radiation field, iterating the spatial solution where required. MoCHII brackets the root of heating minus cooling in $\log T_e$ and re-solves the H/He and metal ionization at every trial temperature. Its current budget includes photoionization heating, H/He recombination and collisional losses, free-free emission with a tabulated Gaunt integral (H/He and, by default, the trace metals with their net ion charge), metal-line cooling, optional metal photoheating, secondary ionization, and the atomic-PDR grain and neutral collision terms.

The MAPPINGS thermal solver used by MAPPINGS V 5.2.1 and inherited by M^3 includes a broader traditional plasma budget: hydrogen and helium line cooling, fine/intercombination/resonance and multilevel metal cooling, free-free, collisional ionization, recombination, photoionization heating, charge-exchange energy exchange, cosmic rays, grain/PAH photoelectric terms, and grain collisional cooling. The microturbulent dissipation block in the examined 5.2.1 `cool.f` is commented out and is therefore not listed as active. This breadth and the direct multilevel coupling are major MAPPINGS strengths.

8. Dust, PAHs, and PDR physics

8.1 Where the MAPPINGS microphysics is stronger

The MAPPINGS grain microphysics is detailed. The local source tree carries graphite and silicate size distributions, depletion bookkeeping, grain charge balance, electron and ion sticking, grain photoelectric heating, collisional grain cooling, PAH charge states, and optional thermal or ionization-parameter-dependent grain destruction. For the local exchange of energy and charge between gas and a resolved grain population, this is more microscopic than the current MoCHII Bakes–Tielens heating prescription.

In MAPPINGS V 5.2.1, this layer is connected to the P6/P7 zone integration and to `dusttemp`, which computes a Draine–Li stochastic temperature distribution and an IR photon field. It is therefore a complete 1D dust emission path, not merely a library capability. The distinction below is specific to the public M^3 three-dimensional branch.

8.2 Where the three-dimensional M^3 dust path is limited

The public `montph7_dust` branch includes dust in the extinction and terminates a packet when the selected interaction is grain absorption. The dust data contain absorption, scattering, and asymmetry information, but the examined 3D branch computes an absorption fraction and does not sample a new dust-scattering direction. The dedicated `dusttemp` routine is called from the one-dimensional `photo6/7` family, not from the public 3D driver. Consequently, the rich MAPPINGS grain microphysics should not be conflated with a complete 3D dust radiative transfer and emergent-IR calculation.

This does not imply that dust is absent from M^3: dust opacity, depletion, grain heating/cooling, and PAH processes affect the gas. It means that the public path does not presently expose a validated chain equivalent to MoCHII's dust scattering, absorbed-energy ledger, stochastic emission, and observer maps.

8.3 The MoCHII dust chain

MoCHII couples dust to the same FUV/EUV radiation field that sets the gas state:

1. Gas and dust compete for ionizing photons using bin-dependent absorption cross sections. The dusty Strömgren gate reproduces the one-dimensional radius to +0.69% and the classic radius reduction $R/R_0 = 0.735$.

2. With scattering enabled, the extinction includes the dust scattering coefficient, a forced first event is sampled, and the packet follows an HG direction with an albedo weight. The dusty-sphere result lies between the pure-absorption and extinction-as-absorption bounds.

3. The live dust model updates the dust abundance from the computed neutral fraction. A separate PAH survival fraction permits destruction inside ionized gas and produces a PAH-bright I-front ring.

4. The absorbed FUV/EUV power gives a leaf equilibrium mixture temperature and an energy-conserving IR spectrum.

5. SEDust supplies astrodust, DL07, or Zubko stochastic emission. The final normalization enforces $\int L_\lambda \, d\lambda = \sum_{\rm leaf} H_{\rm dust} V$; the smoke test closes at unity and resolves the usual PAH bands.

6. Leaf SEDs at selected wavelengths generate spatially resolved 8, 24, 70, and 160-μm maps.

For dusty three-dimensional H II regions, this end-to-end chain is the clearest overall physical advantage of MoCHII.

8.4 PDR scope

MoCHII extends the band below 13.6 eV so that low-threshold metals and dust see the FUV field beyond the H I-front. Optional switches include electrons from the metal cascade, metal photoheating, Bakes–Tielens grain photoelectric heating plus recombination cooling, and neutral-H impact cooling in [C II] 158 μm and [O I] 63 μm. This is an atomic PDR skin, not a complete PDR code. H_2/CO chemistry, molecular self-shielding, molecular cooling, line transfer, and gas–dust collisional coupling remain outside the model.

The MAPPINGS trees likewise contain more grain and general plasma physics than a minimal H II solver, but the present comparison did not identify a full molecular H_2/CO network in either the 5.2.1 photoionization path or the public M^3 3D driver. None of the three codes should be selected as a substitute for a dedicated molecular PDR code without additional chemistry.

9. Emission lines, continua, and synthetic observations

9.1 Line inventories and emissivity accuracy

The MAPPINGS lineage was designed to predict a very broad line spectrum. The M^3 cooling and output path includes resonance, intercombination, forbidden, fine-structure, hydrogenic, helium, heavy recombination, and extensive iron-line data. Its five-level and multilevel populations are already part of the thermal solution. This makes M^3 attractive for integrated diagnostic grids and hard nebulae for which a large fraction of the periodic table and many ion stages matter. MAPPINGS V 5.2.1 is the strongest version of that argument when 1D geometry is acceptable: the full preference set covers 30 elements, and the deterministic zone solver can afford a detailed spectrum without Monte Carlo sampling noise. M^3 trades some parent-code recency and 1D special modes for arbitrary Cartesian geometry.

MoCHII has a narrower but highly auditable diagnostic layer. Every registered ion with a Tier-2 file is solved by the same partial-pivot statistical-equilibrium code using fitted CHIANTI v11 effective collision strengths and explicit A-values. H I and He II recombination lines use Storey–Hummer case-A/B tables. The nebular continuum includes free-bound plus free-free and two-photon components. On an identical state, the Fortran line luminosities match the independent Python solver to at least four significant figures. The limitation is coverage, not the internal emissivity algebra.

9.2 Images and observing geometry

MoCHII writes leaf state and line-emissivity blocks to HDF5/FITS. Its Python reader deposits each AMR leaf with exact pixel-area overlap, so surface-brightness integration recovers the three-dimensional luminosity to machine precision. It also supports slab cuts and EM-, electron-, hydrogen-, or volume-weighted field maps.

For the transported FUV/EUV field, an extra converged-state pass applies observer peel-off. Stellar and diffuse emission contribute direct $e^{-\tau}$ terms, every dust interaction contributes an HG-weighted scattered term, and the optical-depth walk uses the same gas and dust opacity as the rate calculation. Multiple observers, unattenuated direct images, and bin-resolved cubes are available. The direct thin-medium gate matches the analytic image-integrated flux by +0.15% under the current defaults (exact hydrogenic cross sections, threshold-aligned bins).

M^3 outputs cell-resolved physical quantities and line emissivities, and its published applications construct spatial line maps from them. This is adequate for optically thin line projection. The public driver does not show a comparable general next-event estimator for arbitrary observers, nor the separation of direct and dust-scattered continua. Thus M^3 is richer in the number of intrinsic lines, while MoCHII is richer in the chain that turns a three-dimensional state into calibrated synthetic observations.

10. Convergence, verification, and maturity

10.1 Convergence criteria

M^3 updates the radiation field, solves each illuminated cell, and counts cells whose temperature and H^+ changes meet the selected tolerances. The run stops when a requested fraction of ionized cells has converged or after the iteration limit. Its public defaults use percent- level cell changes and increase the packet count when convergence stalls. This is intuitive, but the result is sensitive to how the ionized-cell mask and required fraction are chosen.

MoCHII reports both cell-max and volume-integrated pairs:

$$\delta_x^{(V)} = \frac{\sum V\,|\Delta x_{\rm HII}|}{\sum V x_{\rm HII}}, \qquad \delta_T^{(V)} = \frac{\sum V n_{\rm e}\,|\Delta T_{\rm e}|}{\sum V n_{\rm e} T_{\rm e}}. \tag{4}$$

The volume criterion suppresses front-cell jitter by the small volume of the front and excludes temperature oscillations in nearly electron-free cells. In the FUV dusty test, cell maxima remained at 0.1 and 6.4 while the volume measures reached stable Monte Carlo floors of 2.7×10^{-3} and 6.6×10^{-3}. Thirty further iterations changed the ionized volume by only 0.014%. This is a stronger convergence diagnostic for turbulent, clumpy grids, provided the tolerance is set just above the packet-noise floor.

10.2 Validation evidence

The MAPPINGS lineage has the stronger external maturity record. M^3 inherits decades of MAPPINGS development and has been published against the HII20, HII40, and PN150 Lexington/Meudon standards. Its three-dimensional blister, bipolar, and fractal applications demonstrate the importance of geometry for spatially resolved line diagnostics [1, 2].

MoCHII is newer, but its current repository contains unusually granular gates:

- analytic H/He attenuation and rate integrals;
- a Strömgren sphere against an exact one-dimensional reference;
- AMR versus uniform and solution-driven refinement versus native high resolution;
- H/He thermal balance against an independent Gauss–Seidel reference;
- direct CHIANTI fit evaluation and PyNeb line-ratio checks;
- MOCASSIN HII20/HII40 state and line comparisons;

- case-A/case-B diffuse-field bracketing;
- dust absorption/scattering brackets and IR energy closure;
- direct peel-off flux against an analytic observer solution; and
- exact three-dimensional emissivity-to-map luminosity conservation.

These gates provide strong internal evidence and make regressions easy to localize. They do not yet replace the independent user base and long publication history of MAPPINGS. The correct conclusion is that the MAPPINGS lineage is more mature externally, while MOCHII is more explicit and fine-grained in its current numerical audit trail.
MAPPINGS V 5.2.1 does not have a Monte Carlo convergence problem: zone step control and the global shock/precursor iteration dominate instead. Its decades of use and broad model suite are a major maturity advantage, but the inspected 5.2.1 tree does not provide a gate hierarchy as explicit as the current MOCHII test directories. A controlled local benchmark is still required before attributing any line difference to geometry rather than to the 5.1.13-versus-5.2.1 atomic and solver divergence.

11. Software architecture and maintainability

MOCHII uses Fortran modules, allocatable leaf arrays, a single namelist, MPI-3 shared memory, HDF5/FITS sections, and a data-driven species registry. New atomic products are generated offline by Python tools and carry machine-readable provenance. A new ion normally requires no new solver code. The transport, ion balance, thermal balance, lines, dust, and output layers are separated modules. This architecture substantially reduces the cost and risk of adding physics.
The public M^3 tree retains the Fortran 77 MAPPINGS architecture: large common blocks, compile-time maxima, interactive input, and many global arrays. Several historical `montph7_*` variants duplicate large parts of the transport driver. This structure has the advantage of preserving a mature code base and its full menu of MAPPINGS models, but it makes a change to the 3D algorithm harder to propagate and test. The available Sphinx input/output documentation is sparse, so reliable use often requires reading the source and reproducing the authors' setup.
The architectural verdict is therefore clear: MOCHII is easier to extend, audit, automate, and connect to simulation data. M^3's value lies primarily in the depth of the inherited physical library, not in modern software structure.
MAPPINGS V 5.2.1 shares the same common-block and interactive heritage, but its organization is cleaner than the collection of duplicated M^3 `montph7_*` experiments and it has a single public `map52` model selector. Runtime preference files make the 16- or 30-element atom sets configurable without recompilation. Nevertheless, automated model grids require scripting interactive input and parsing many ASCII outputs; the current documentation overview explicitly remains incomplete. For maintainability the ordering is MOCHII first, MAPPINGS V 5.2.1 second, and the experimental M^3 3D fork third.

12. Suitability by science case

Table 3: Recommended code by present science requirement.

Science requirement	Preferred code	Reason
Clumpy or porous H II region from a TNG/RAMSES snapshot	MoCHII	AMR, front refinement, and simulation-oriented I/O.
Ionization-front position, shadows, and clump skins	MoCHII	Adaptive resolution and the analytic absorption estimator.
Dusty Strömgren structure and UV attenuation	MoCHII	Gas–dust competition and validated HG scattering in the 3D band.
PAH rings, IR SEDs, or 8–160-μm maps	MoCHII	SEDust, leaf dust emissivities, and exact absorbed-energy normalization.
Spatially resolved IFU line maps	Usually MoCHII	AMR emissivities and flux-conserving projection; verify that the required ions exist.
Very broad integrated optical/UV/IR line inventory	MAPPINGS V 5.2.1	Current full 30-element MAPPINGS preference set and deterministic detailed spectrum.
Compact high-density nebula	MAPPINGS V 5.2.1 if 1D; M^3 if 3D	Both feed density-dependent multilevel cooling back into T_e; geometry decides.
Hard PN-like ionizing spectrum	MAPPINGS V 5.2.1 if 1D; M^3 if 3D	More high charge states; M^3 additionally has published PN150 3D validation.
Non-Maxwellian κ electron experiment	MAPPINGS V 5.2.1 or M^3	Both MAPPINGS-derived trees have native κ rate support.
One-dimensional radiation-pressure-confined nebula	MAPPINGS V 5.2.1	Isobaric photoionization can include the accumulated radiation pressure.
Finite source lifetime or source-off ionization history in 1D	MAPPINGS V 5.2.1	P6/P7 connect finite-age modes to the time-dependent ion solver.
Steady 1D radiative MHD shock and precursor	MAPPINGS V 5.2.1	S5 couples adaptive post-shock cooling to an iterated auto-preionization solution.
CIE, NEQ, or PIE cooling-curve grid	MAPPINGS V 5.2.1	Dedicated CC, NC, and PP drivers and a broad general-plasma ion inventory.

Table 3 continued.

Science requirement	Preferred code	Reason
Controlled test of modern CHIANTI v11 coefficients	MoCHII	Versioned fitted data and direct fit-error gates.
Three-dimensional time-dependent ionization	None	Both public 3D drivers are equilibrium solvers; MAPPINGS V 5.2.1 time dependence is 1D.
Three-dimensional radiative shock	None	MAPPINGS V 5.2.1 has a strong 1D shock mode, but it is not coupled to either 3D transport architecture.
Molecular H_2/CO PDR	None	All three need a dedicated molecular chemistry and line-transfer layer.
Lyα or metal resonance-line profiles	None	Requires frequency-redistributing resonant transfer, e.g. LaRT.

13. Recommended development path for MoCHII

The comparison suggests a focused roadmap rather than wholesale replacement of either code.

1. **Put density dependence into the thermal cooling.** Reuse the generic n-level solver in the thermal loop for dominant coolants, or fit a suppression factor versus (T_e, n_e) around each critical density. This closes the most consequential gap relative to MAPPINGS.

2. **Extend the registry upward in ion stage.** Add the ions needed for PN and hard-field applications before adding marginal new low stages. The existing data-operation design is well suited to this.

3. **Add Na, Al, and Ni only when driven by a diagnostic need.** Their MAPPINGS implementations provide useful reference behavior, while the MoCHII pipeline should retain explicit provenance and ion-by-ion gates.

4. **Retain the present transport architecture.** The octree, analytic estimator, Cartesian DDA, threshold-aligned bins, and peel-off layer are major advantages and should not be replaced by the legacy M^3 packet synchronization model.

5. **Use MAPPINGS V 5.2.1 as the differential microphysics oracle.** Construct one-zone tests at fixed (T_e, n_e, J_ν) and compare ion fractions, heating/cooling terms, and individual line emissivities before comparing full 3D nebulae. Use M^3 only as the second-stage test of how that inherited microphysics behaves in arbitrary geometry.

6. **Perform a controlled common-physics benchmark.** Use the same uniform grid, source SED, abundances, dust-off setting, frequency range, and atomic subset. Compare J_ν, photo-rates, ion fractions, T_e, cooling budgets, and

line emissivities separately. Only then compare total runtime and MPI scaling.

The most valuable hybrid would combine MAPPINGS V 5.2.1-like density-dependent microphysics with the current MoCHII geometry, transport, dust, and observer layers. This would preserve the strongest feature of each code without inheriting the principal numerical and software limitations of the other.

14. Final assessment

Capability	Stronger now	Confidence
Three-dimensional transport estimator	MoCHII	High
Adaptive spatial resolution	MoCHII	High
Dust scattering and IR/PAH observables	MoCHII	High
Observer imaging and data products	MoCHII	High
Atomic-data provenance and fit auditability	MoCHII	High
Element and ion-stage breadth	MAPPINGS V 5.2.1	High
Density-dependent thermal line cooling	MAPPINGS V 5.2.1/M^3	High
Hard PN/general plasma coverage	MAPPINGS V 5.2.1	High
Finite-age 1D photoionization	MAPPINGS V 5.2.1	High
Radiative MHD shock and precursor	MAPPINGS V 5.2.1	High
Noise-free 1D parameter-grid throughput	MAPPINGS V 5.2.1	High
External validation history	MAPPINGS lineage	High
Software extensibility and automation	MoCHII	High
Absolute line accuracy for every ion	Case dependent	Requires a common-data benchmark
Massively distributed memory ceiling	Architecture dependent	Requires a scaling experiment

For the principal intended application–dusty, structured, spatially resolved H II regions–MoCHII is the better current platform. For broad integrated emission-line diagnostics, dense nebulae, harder ionization conditions, finite-age 1D photoionization, and radiative shocks, MAPPINGS V 5.2.1 is the more complete physical reference. M^3 becomes preferable only when those MAPPINGS-like line and cooling strengths must be combined with arbitrary three-dimensional Cartesian geometry. None of these judgments is universal: first decide whether geometry, time history, shock dynamics, dust observables, density-dependent cooling, or ion-stage breadth is the controlling requirement.

15. Hard-spectrum (PN and AGN) extension plan

This section records the staged plan that follows from the comparison (kept as `docs/HARD_SPECTRUM_PLAN.md`, drafted 2026-07-18, not yet started). It uses the same staged-gate discipline as `docs/PLAN.md`.

15.1 Scope and the two tracks

"Hard spectrum" splits into two physically distinct regimes, kept as separate tracks because they need different physics:

- **PN track** — thermal central stars, $T_{\rm eff} \sim 75$–200 kK. A 150 kK blackbody has $kT = 12.9$ eV; its photon flux at 300 eV sits $\sim 23\,kT$ out on the Wien tail ($e^{-23} \sim 10^{-10}$ before the phase-space factors), and the lowest K-shell edge of an abundant metal (C, 290 eV) is never reached with significant flux. The MOCASSIN PN150 benchmark itself integrates its band only to `nuMax` = 24 Ryd = 326 eV. The PN track therefore needs *no inner-shell/Auger physics and no Compton terms* — it is band headroom, higher ion stages, the He II diffuse channels, and (at PN densities) density-dependent cooling.

- **AGN track** — a power-law/big-bump continuum extending to keV X-rays. Above $\sim$0.3 keV inner-shell absorption dominates the metal opacity and Auger multi-electron ejection couples non-adjacent ion stages; near $\sim$0.5–1 keV the Thomson cross section (6.65×10^{-25} cm^2) overtakes the steeply falling ($\sigma \propto E^{-3}$) photoabsorption of the residual neutrals, and Compton heating enters the energy budget. The first increment caps the band at $\sim$500 eV, where photoabsorption still dominates, and defers the Compton terms (Section 15.10).

Current baseline (v1.00-dev, 2026-07-18): band 13.6–100 eV (+FUV option); registry stages C 4 / N 3 / O 3 / Ne 3 / S 4 / Ar 4 / Mg 3 / Fe 4 / Si 5 / Cl 5 / Ca 5, with the compile-time stage ceiling `MAX_ST` = 6 in `species_mod` — the PN targets below fit it exactly, so no framework change is needed there; `te_max` = 5×10^4 K and the Tier-1 cooling fits are generated over 10^3–10^5 K; $\bar{g}_{\rm ff}$ is tabulated for $Z_{\rm ion}$ = 1–4 (`cooling_mod`, parameter `NGZ`); secondary ionization (`use_sec_ion`, Shull–van Steenberg 1985) is implemented but default off; the Owen-scrambled Sobol launch is available. The He II Storey–Hummer lines (case A and B), the He I Porter lines, and the diffuse He III ground channel already cover the He II 4686 diagnostic side, and the aligned-edge band builder (`ion_align_edges`, default on) with its graceful threshold-dropping logic is exactly the piece that makes many new in-band edges safe.

15.2 Reference assets (verified present)

Table 4: Local reference assets for the hard-spectrum work.

Asset	Use
MOCASSIN PN150 + PN75 benchmark inputs (`benchmarks/gas/{PN150,PN75}`)	primary 3D Monte Carlo gates; run MOCASSIN itself *converged* (the HII40 lesson). PN150 input read directly: 150 kK blackbody, $n_{\mathrm{H}} = 3000$ cm^{-3}, $R_{\mathrm{in}} = 10^{17}$, $R_{\mathrm{out}} = 4.07 \times 10^{17}$ cm, band to 24 Ryd, abundances He 0.10, C 3×10^{-4}, N 10^{-4}, O 6×10^{-4}, Ne 1.5×10^{-4}, Mg 3×10^{-5}, Si 3×10^{-5}, S 1.5×10^{-5} (`abunPN150.in`) — note Mg and Si ARE in the composition, which drives their stage targets below.
Cloudy c23.01 test suite (`tsuite/auto/pn_paris*.in`, `nlr_paris.in`, `agn_lex00_u{0,1,m1}.in`)	1D cross-code gates: Meudon PN, AGN NLR, and the Lexington-2000 X-ray slabs. The `agn_lex00` inputs read directly: $n_{\mathrm{H}} = 10^5$ cm^{-3}, $\phi(\mathrm{H}) = 10^{15.477}$ cm^{-2}s^{-1}, stop column 10^{16} cm^{-2}, and a *tabulated* SED (Cloudy `interpolate/continue` points) — so MoCHII reproduces the illumination exactly through the existing `spectrum_type` file route, without waiting for the AGN preset.
Cloudy AGN SED (`source/parse_agn.cpp`)	the standard AGN continuum parameterization (big-bump T, α_{ox} anchored at 2500 Å–2 keV, α_{uv}, α_{x}) for a MoCHII preset.
Kaastra & Mewe (1993) Auger yields (`data/mewe_nelectron.dat`, `mewe_fluor.dat`)	electrons ejected per inner-shell vacancy, shell-resolved, machine-readable provenance for the Auger stage.
Verner & Yakovlev (1995) shell-resolved cross sections (`sigma_vy95` already in `photo_xsec.f90`)	inner-shell photoionization above the K/L edges.
CHIANTI v11.0.2 + Badnell 2023 RR/DR (in-tree)	levels/collision strengths and recombination for every new ion stage (coverage verified).

15.3 HS1: band headroom

Mostly validation; the code changes are one-line parameters.

1. `eion_max` (define.f90, default 100 eV) raised to 300–500 eV for hard runs. This matters even for stages the registry ALREADY tracks: the Ne IV$\rightarrow$V threshold is 97.1 eV — just under the present cap — so today's band cannot photoionize to Ne V at all, and the new O/Ne thresholds (113.9, 126.2 eV) lie beyond it. Cross-section validity is not the issue (the Verner et al. fits extend to keV energies and H I/He II are exact hydrogenic); the work is verifying the pieces that ASSUME the band shape: the aligned-edge builder with $\sim$30–40 in-band thresholds once HS2 lands

(recommend `nnu_ion` ≥ 64, 128 clean; the largest-remainder allocator and the metal-edge dropping logic already degrade gracefully), the FUV+ionizing two-segment bookkeeping, the diffuse `ion_bin_of` edge search, and the secondary-ionization band integrals whose $E_0 > 40$ eV branch becomes the COMMON case on a hard field.

2. $\bar{g}_{\rm ff}$: raise `NGZ` from 4 to 8 in `cooling_mod` (the van Hoof table already spans the required γ^2 range; the metal free-free sum then covers the new stages).

3. `te_max`: allow 10^5–10^6 K for X-ray-heated zones. Any ion used above 10^5 K needs its Tier-1 fit regenerated over the wider range (a flag in `chianti_cooling.py`, not new code); the bisection itself is range-agnostic.

4. Recommended hard-field defaults, documented in the example inputs: `use_sec_ion=.true.` (a 150 kK field makes $E_0 > 40$ eV photoelectrons routine, and X-ray fields more so) and `launch_sequence='sobol'` (the shielded, partially neutral zones where secondaries act are exactly where launch noise hurts most).

Gate HS1: the G0 pattern on the widened band — fixed-state analytic attenuation with a 150 kK blackbody, $\Gamma_{\rm HI/HeI/HeII}$ vs the analytic integrals bin by bin; $Q(>13.6)/Q(>24.6)/Q(>54.4)$ photon-rate ratios against exact Planck integrals to $< 0.1\%$ with aligned edges; and a spot check that the tabulated VFKY96 metal cross sections at 300–500 eV match the published fit values (guards against table-range assumptions in `photo_xsec`).

15.4 HS2: registry upward in ion stage

A data operation through the existing fitting pipeline:

Table 5: Target stages and the diagnostics they unlock.

El.	now→target	new thresholds [eV]	key new lines
C	4→5	C IV→V 64.5	C IV 1548/1550 (resonance — optically thin caveat)
N	3→5	47.4, 77.5	N IV] 1486, N V 1240
O	3→6	54.9, 77.4, 113.9	[O IV] 25.9 μm, O IV] 1400, O V] 1218
Ne	3→6	63.4, 97.1, 126.2	[Ne IV] 2422/2425, [Ne V] 3346/3426 + 14.3/24.3 μm, [Ne VI] 7.65 μm
S	4→6	47.2, 72.6	[S IV] 10.5 μm (new *n*-level file)
Ar	4→6	59.8, 75.0	[Ar V] 7.90/13.1 μm, [Ar VI] 4.53 μm
Mg	3→6	80.1, 109.3, 141.3	stage tracking for the PN150 composition (Mg 3×10^{-5}); [Mg IV] 4.49 μm, [Mg V] 5.61 μm as data permit
Si	5→6	Si V→VI 166.8	stage tracking for the PN150 composition (Si 3×10^{-5})

Fe/Cl/Ca hold (extend only on diagnostic demand; coronal [Fe VII+] deferred). All targets fit the existing `MAX_ST` = 6 ceiling exactly; `MAXLEV` = 40 / `MAXTR` = 600 and the 600-line `lines_mod` arrays already accommodate the new n-level models. Work items, all through `tools/fitting/`:

1. **Element files** (`make_element_data.py`): VFKY96 `PHOTO` rows exist in phfit2 for every target stage; Voronov CI covers all stages; Badnell 2023 RR+DR cover these isoelectronic sequences in the in-tree `clist_K` files (with the CHIANTI `RR2`/`DR2` fallback forms already supported where a fit is absent). **Charge exchange** above stage ∼4 is not in the Kingdon–Ferland fits: adopt the fast Dalgarno-type $\sim 1.92 \times 10^{-9} z\ \mathrm{cm^3 s^{-1}}$ recombination-only scaling used by Cloudy, with an explicit provenance note in each element file, or document neglect — decide once, apply uniformly (CX matters little in the hot interior where these stages live, but the decision must be recorded).

2. **Tier-1 cooling fits** (`chianti_cooling.py`) for each new stage, fit range matched to the chosen `te_max`.

3. **Tier-2 n-level files** for the diagnostic ions ([Ne IV], [Ne V], [O IV], [S IV], [Ar V], [Ar VI], N IV, N V, C IV); CHIANTI v11 carries level and collision data for all of them. Ions without CHIANTI level data are tracked without lines — the established Ar II pattern.

4. **PN abundance example** (`examples/`): the `abunPN150.in` values verbatim, so the PN150 gate input is reproducible from the repository alone.

Gate HS2: PyNeb ratio checks ion by ion at the established database-spread tolerances (sub-percent for well-determined optical pairs, up to ∼5–10% for the known compilation-spread classes): the [Ne V] 3426/24.3 μm temperature pair, the [Ne IV] 2422/2425 and [Ar IV] 4711/4740 density pairs, [O IV] 25.9 μm against its A-value limit; then an end-to-end hard-field smoke showing every new line in `_lines.txt` with physically sane ratios.

15.5 HS3: He II excited-channel diffuse photons

The PN He III zone is large, and case-B He II recombinations emit ionizing photons beyond the ground continuum already carried (`diffuse_mod` channel 3). The hydrogenic $n = 2$ cascade of He II produces three components, every one of which ionizes H I:

- He II Lyα at 40.8 eV — ionizes H I AND He I, the dominant piece;
- the He II Balmer continuum — its threshold is $Z^2\,\mathrm{Ry}/n^2 = 13.60$ eV, i.e. exactly at the H I edge; the aligned-grid `ion_bin_of` binary search (the 2026-07-18 bug fix) is precisely what makes this land in the correct bin;
- the He II two-photon continuum (0–40.8 eV), of which the > 13.6 eV part ionizes — sampled like the He I two-photon branch with its ionizing probability precomputed.

Implementation is the `hei_diffuse` pattern verbatim: a fifth channel in `dif_ch` weighted by the case-B He III recombination rate in each leaf, branch fractions from the hydrogenic 2s/2p cascade shares (Osterbrock & Ferland tables), energies fixed (Lyα), threshold-plus-exponential (Balmer continuum), or flat-sampled (two-photon). The quasi-random diffuse launch needs NO new dimensions — the channel pick d7 simply spans five values. Regime note: at HII40 the analogous He I channel moved T_e by only ∼1%, but there the He III zone was a small core; in a 150 kK PN the He III region fills much of the nebula and this becomes a first-order energy-budget term.

Gate HS3: channel energy conservation (emitted diffuse luminosity equals the case-B $n \geq 2$ recombination power of the He III zone); the PN150 He structure and T_e recorded with the channel off/on, attributed as one lever.

15.6 HS4: PN validation gates

MOCASSIN PN150/PN75, with the full-quality protocol that resolved HII40: MOCASSIN re-run CONVERGED (its stored reduced settings — 10^6 photons, 10 iterations, `convLimit` 0.05, 13^3 — are a moving target exactly as the HII20/40 baselines were), and MoCHII run with `conv_crit='vol'`, `nnu_ion` 64–128 aligned, $\geq 4 \times 10^7$ packets, `launch_sequence='sobol'`. The PN150 setup is fully specified by the benchmark input quoted in Table 4 (150 kK, $n_H = 3000$ cm^{-3}, shell 10^{17}–4.07×10^{17} cm). Compare, in order of diagnostic power:

1. T_e(H II) and the radial T_e profile (PN Te $\sim$12–18 kK — also exercises the hotter thermal range);
2. the He I/II/III structure — the He II-edge (54.4 eV) photon budget is the PN analog of the He edge discretization that dominated HII40, so the aligned grid must show its value here;
3. the new high-stage fractions O III/IV/V, Ne III/IV/V, N III/IV;
4. the line table: L(Hβ), He II 4686/Hβ, [O III] 5007/4363, [Ne V] 3426, [Ne III] 15.5 μm, [O IV] 25.9 μm, [N II] 6584, [S IV] 10.5 μm, [Ar V].

Cloudy `pn_paris` as the 1D cross-check with identical SED, abundances, density, and dust off: zone-by-zone T_e and ion fractions along the radius plus $\sim$20 integrated ratios. Because Cloudy solves density-dependent level populations in its thermal loop, the PN150 residual BEFORE stage HS7 directly measures how much the low-density Tier-1 fits miss at $n_H = 3000$ cm^{-3} — that number is the input to HS7, which is why HS4 deliberately precedes it.
Acceptance: land inside (or bracket) the published Lexington PN150 cross-code spread, with each residual attributed one lever at a time (binning, diffuse channels, density-dependent cooling, data vintage) exactly as the HII40 campaign did.

15.7 HS5: AGN SED input

An `ion_spectrum='agn'` preset implementing the Cloudy continuum (`parse_agn.cpp`; Mathews–Ferland form):

$$f_\nu = \nu^{\alpha_{uv}} e^{-h\nu/kT_{bump}} e^{-kT_{IR}/h\nu} + a\nu^{\alpha_x}, \tag{5}$$

with the X-ray normalization a fixed by α_{ox} through the 2500 Å–2 keV energy ratio (the 403.3 factor in the Cloudy source) and the standard defaults $\alpha_{uv} = -0.5$, $\alpha_x = -1$, IR cutoff at $kT_{IR} = 0.01$ Ryd. It enters `ion_band_mod` exactly like the Planck function — a spectral density integrated on the 32-point sub-grid in each band bin — and is normalized by the existing derive-or-rescale luminosity rules. The tabulated-file route (`spectrum_type='per_ev'` etc.) remains the reference path and already reproduces the `agn_lex00` gate SEDs verbatim (Table 4); the preset adds convenience and reproducibility for parameter studies. Also add an ionization-parameter helper to the rank-0 log, $U = Q_{ion}/(4\pi r^2 n_H c)$ evaluated at a reference radius, so slab setups map onto Cloudy inputs without hand conversion.
Gate HS5: the bin-integrated preset spectrum against Cloudy `save continuum` output for identical parameters (shape agreement to $\sim$1% per bin); Q-ratio integrals exact; the derive/rescale bookkeeping re-verified on the preset (the ISRF-preset test pattern).

15.8 HS6: inner-shell absorption and Auger (AGN track)

The one genuine framework extension of the plan, in four parts:

1. **Shell-resolved opacity.** Above each K/L edge the total cross section becomes the VFKY96 outer-shell fit plus the Verner & Yakovlev (1995) inner-shell partials, $\sigma_{\rm tot}(E) = \sigma_{\rm outer}(E) + \sum_s \sigma_s(E)$. The `sigma_vy95` evaluator and the `PHOTO2` element-file rows already exist (the Cl path); the extension is carrying several shells with their edges in the aligned-band threshold list.

2. **Auger yields.** Kaastra & Mewe (1993) electrons per vacancy, shell-resolved, read from the Cloudy data file format (`mewe_nelectron.dat`: element, ion, shell, yield distribution): an inner-shell ionization of stage i lands at stage $i+n_{\rm e}$(shell) with the tabulated probability distribution over $n_{\rm e}$. Fluorescence (Kα photon emission instead of one Auger electron, `mewe_fluor.dat`) is a small branch for low-Z elements; the first increment deposits it as local heat with a documented note rather than transporting the line photon.

3. **Cascade generalization.** `species_fractions` currently solves the strict adjacent-stage chain as a product form. Auger couples stage i to $i+2, i+3, \ldots$, so each element's stationary balance becomes a small $n_{\rm stage} \times n_{\rm stage}$ matrix problem ($M\mathbf{x} = 0$ with $\sum_j x_j = 1$); the partial-pivot elimination already used by `nlevel_mod` is reused directly. The chain path is KEPT as the default and the matrix path activates only when an inner-shell channel is present — the off-path stays arithmetically identical, the house rule for every extension.

4. **Energetics.** The Auger electron carries $E_0 = h\nu - E_{\rm th,shell} - \sum E_{\rm secondary\ vacancies}$ and is fast; it enters the EXISTING Shull–van Steenberg partition through the absorber-resolved excess-energy integrals with no new heating physics. Compton remains excluded at $\leq$500 eV.

Gate HS6: the X-ray-illuminated slab against the Cloudy `agn_lex00_u{m1,0,1}` set (three ionization parameters at $n_{\rm H} = 10^5$ cm^{-3}, tabulated SED, stop column 10^{16} cm^{-2}): ion-fraction profiles of C/N/O/Ne through the partially ionized zone, the $T_{\rm e}$ profile, and the H/He recombination lines, within the published Lexington-2000 cross-code spread. A secondary check: with Auger artificially disabled the slab must reproduce the HS2-era solution (isolates the new coupling).

15.9 HS7: density-dependent metal cooling in the thermal loop

Required at PN densities. The coolants that dominate at nebular temperatures quench in a specific order because their critical densities span five decades (approximate values at 10^4 K): [C II] 158 μm $n_{\rm crit,e} \sim 50$, [N II] 122 μm $\sim 3 \times 10^2$, [O III] 88 μm $\sim 5 \times 10^2$, [O III] 52 μm $\sim 3.5 \times 10^3$, [O II] 3729 $\sim 3 \times 10^3$, [S II] 6716 $\sim 2 \times 10^3$, [N II] 6584 $\sim 9 \times 10^4$, [O III] 5007 $\sim 7 \times 10^5$. At the PN150 density ($n_{\rm H} = 3000$, $n_{\rm e}$ up to $\sim$7000 cm^{-3} in the He III zone) the IR fine-structure lines are already partly quenched while the optical forbidden lines are not — the low-density Tier-1 fits therefore OVERESTIMATE cooling and bias $T_{\rm e}$ low, with the error growing toward compact PN and NLR densities.
Primary design, keeping "fitted offline, evaluated online":

1. `chianti_cooling.py` already solves the full n-level system to build its fits; extend it to emit $\Lambda(T, n_{\rm e})$ tables on a log-$n_{\rm e}$ grid ($\sim$7 nodes, 10^1–10^7 cm^{-3}) $\times$ the existing log-T range, one block appended to each `cooling_tier1_*.txt`

behind a format-version flag (old files keep working).

2. `species_mod::metal_cooling` evaluates bilinear-in-($\log T, \log n_\mathrm{e}$); the memory cost is trivial ($\sim 60 \times 7$ values for each ion) and the evaluation stays a handful of multiply-adds inside the bisection — the `species_ne` cache pattern applies unchanged.

3. The lowest-density node MUST reproduce the current fits to fit accuracy (regression guard), so HII20/HII40 and every recorded gate move by $< 0.1\%$.

4. Verification: an in-loop n-level solve for the dominant coolants on a small model, confirming table interpolation error $< 1\%$ in Λ over the used (T, n_e) range; the H-impact PDR cooling terms (`metal_cooling_H`) stay as they are (their two-level low-density form is correct in the PDR regime they serve).

Gate HS7: a two-zone analytic check (one low- and one high-density zone against the exact n-level cooling); PN150 T_e with the tables off/on (the residual measured at HS4 must shrink accordingly); HII20/HII40 unchanged to $< 0.1\%$.

15.10 Deferred items

Recorded once so they are conscious exclusions, not oversights: Compton heating/cooling and Thomson-scattering transport ($\gtrsim$0.5 keV; Cloudy is the reference implementation); Fe K fluorescence photon transport (the yield enters HS6 as local heat); coronal ions beyond $\sim$stage 8 ([Fe VII] 6087, [Fe X] 6374 need Fe stages 8–11 and a hotter thermal range); X-ray and UV resonance-line transfer (LaRT territory); grain destruction/sputtering in hard radiation fields (the PAH survival switches are a partial hook, not a destruction model); and time dependence / finite source age, where MAPPINGS remains the 1D reference.

15.11 Suggested order

Order	Stage	Framework risk	Blocking
1	HS1 band headroom	low	gates HS2+
2	HS2 registry upward	none (data)	gates HS4
3	HS3 He II diffuse channels	low	sharpens HS4
4	HS4 PN gates (MOCASSIN + Cloudy)	none	PN track done
5	HS7 density-dependent cooling	medium	PN accuracy
6	HS5 AGN SED preset	low	gates HS6
7	HS6 inner-shell/Auger	high (cascade)	AGN track done

HS7 can run in parallel with HS2–HS4 (independent code paths); the first PN150 pass deliberately precedes HS7, so it measures how much of the PN residual the density-dependent cooling must explain — the same one-lever-at-a-time attribution that worked for HII40.

16. Feasibility: time dependence, cooling flows, and shocks in MOCHII

Section 16. addresses the three capabilities where the public M^3 3D driver and MOCHII both trail MAPPINGS V 5.2.1: the connected time-dependent ionization history of P6/P7, the 1D cooling flow, and the S5 steady radiative shock with an iterated precursor (§3.2). The question is not whether these are useful — they plainly are — but how much each would cost to add to MOCHII, given its architecture. The three answers differ sharply: time dependence is a bounded project that the design accommodates naturally, a cooling flow rides on top of it, and a radiative shock is a separate one-dimensional radiation-hydrodynamic calculation that the Monte Carlo transport engine does little to help.

16.1 The architectural pivot

MOCHII's iteration loop transports Monte Carlo packets, tallies the cell radiation field, solves the *equilibrium* ionization and thermal balance in each cell (`solve_ion_cell`: a damped fixed point on n_e with a bisection on T_e), refreshes the opacity, and repeats until converged. The state solver is therefore a *root finder*: given the rates it returns the steady state. It integrates no dn_i/dt, carries no velocity field, and solves no fluid equations. That single fact sets the cost of each addition: one reuses the root finder almost unchanged, one adds a time integrator beside it, and one needs what the code does not have at all.

16.2 Time-dependent ionization: moderate, and the natural fit

Almost everything a finite-lifetime calculation needs already exists. The transport, the photoionization, recombination, collisional-ionization, and charge-exchange rate coefficients (already evaluated cell by cell), the tally, the octree, and the diffuse field are all reusable without change. The cell solver is already "given the rates, find the state"; the new piece is a sibling routine, "given the rates, the old state, and Δt, find the new state."

What is genuinely new is threefold. First, a stiff ordinary-differential integrator for the ionic populations and, where the thermal time is not short compared with the ionization time, the electron temperature — a backward-Euler or semi-implicit step in each cell, of the kind MAPPINGS V 5.2.1's `timion` performs, rather than the present root find. Second, an outer time loop with a step controller keyed to the atomic, cooling, and photon-absorption times. Third, a source light curve or a finite on/off lifetime as input.

The design accommodates this cleanly through operator splitting. MOCHII already performs one transport pass for each iteration, so the natural structure is an outer radiation step that holds J_ν fixed while the chemistry is substepped in each cell, with the radiation field recomputed once the ionization has advanced enough to change the opacity. An ionization front that advances while a source brightens, or recedes after it switches off, is a reachable target. The hard part is the stiff integrator and the step control, not the infrastructure.

16.3 Cooling flows: depends on which cooling flow

The term carries two meanings, and they price very differently.

In the MAPPINGS V 5.2.1 sense — a parcel of gas that cools radiatively as it moves down a prescribed density and velocity history — a cooling flow is a special case of time-dependent cooling on a fixed trajectory: switch off (or freeze) the photoionizing field, and integrate the non-equilibrium ionization and the radiative losses down a temperature ramp. Once the time-dependent solver of the previous subsection exists, this is a modest addition, because the cooling terms it needs — recombination, free–free, and the metal-line coolants through the Tier-1 fits and the Tier-2 n-level solve — are already present. Its one real prerequisite is the density-dependent cooling flagged as HS7 and in §7.4: a cooling flow crosses coolant critical densities, so a $\Lambda(T_e, n_e)$ that suppresses forbidden lines correctly is required for it to be quantitative.

In the cluster or galaxy sense — a self-consistent inflow driven by its own radiative losses, with gravity and hydrodynamics — a cooling flow is a fluid-dynamics problem and lies outside the present scope without a hydrodynamic solver. The comparison with MAPPINGS V 5.2.1 concerns the first meaning.

16.4 Radiative shocks: hard, and off the architecture

The MAPPINGS V 5.2.1 S5 shock solves the Rankine–Hugoniot jump conditions and then integrates the post-shock cooling zone as a one-dimensional flow — mass, momentum, and energy conservation along the flow coordinate with radiative losses — while iterating an upstream photoionized precursor computed from the shock's own radiation to self-consistency. Its essence is a one-dimensional radiation-hydrodynamic flow, not a three-dimensional transport problem.

This is the poor fit. MoCHII has no velocity field, no fluid equations, and no flow coordinate. The post-shock structure is gas that moves and compresses; the static octree grid and the Monte Carlo photon transport represent none of that, and three-dimensional Monte Carlo transport buys almost nothing for a one-dimensional steady shock. The honest route is a separate one-dimensional radiative-shock module, a companion solver in the spirit of the 1D references MoCHII already uses for its gates: it would reuse the atomic and cooling data but not the octree or the Monte Carlo engine, and it would carry the jump conditions, an adaptive cooling-zone integrator, and the precursor iteration itself. That is real work, largely decoupled from the transport code, and — this is the key qualification — it would be a 1D shock, not a 3D radiative shock. A genuine three-dimensional radiative shock needs actual radiation hydrodynamics, which is a different code, not an extension of this one.

16.5 Summary and suggested order

Capability	Difficulty	Why
Time-dependent ionization	Moderate	Reuses all transport and rates; new work is a stiff cell ODE, a time loop, and a light curve. Operator splitting maps onto the existing one-transport-for-each-iteration loop.
Cooling flow (MAPPINGS V 5.2.1 sense)	Moderate	A prescribed-trajectory special case of the time-dependent solver; needs the density-dependent $\Lambda(T_\mathrm{e}, n_\mathrm{e})$ of HS7.
Cooling flow (cluster sense)	Out of scope	Requires gravity and hydrodynamics.
Radiative shock (S5)	Hard	A separate 1D radiation-hydrodynamic module; the octree and Monte Carlo engine do not apply. Not a 3D shock.

A sensible order is: (1) time-dependent ionization first, as the most natural addition and the foundation for the rest; (2) the density-dependent cooling of HS7, needed for both PN accuracy and cooling flows; (3) the MAPPINGS V 5.2.1-sense cooling flow as a thin layer on top of (1) and (2); and (4) a decoupled 1D radiative-shock module last, and only if a shock diagnostic is actually required. The first three share the same time-integration and cooling infrastructure and reinforce the PN and hard-spectrum work of §15.; the fourth stands apart.

17. Implementation evidence consulted

The principal local sources used for this audit were:

- MoCHII: `docs/PLAN.md`, `docs/MoCHII_physics.tex`, `docs/MoCHII_UserGuide.tex`, the transport/grid modules under `src/`, and the quantitative gates under `tests/`.
- M^3: `README.md`, `lab/data/ATDAT.txt`, `src/mastercode/montph7.f`, `montph7_dust.f`, `crosssections.f`, `mcteequi.f`, `teequi2.f`, `equion.f`, `cool.f`, `multilevel.f`, `grainpar.f`, `hgrains.f`, `hpahs.f`, `dusttemp.f`, `timion.f`, and `shock2--shock5.f`.
- MAPPINGS V 5.2.1: `README.md`, `src/mappings.f`, `photo6.f`, `photo7.f`, `teequi.f`, `teequi2.f`, `timion.f`, `timtqui.f`, `cool.f`, `dusttemp.f`, `shock5.f`, `neqc.f`, `coolc.f`, `transferline.f`, `lab/mapStd.prefs`, `lab/mapFull.prefs`, and `docs/DocumentionOverview.html`.

The public M^3 source contains several historical or experimental driver variants. Statements in this memo refer to the main `mastercode/montph7.f` and explicitly selected `montph7_dust.f` paths unless otherwise stated. Statements about MAPPINGS V 5.2.1 refer to the connected public 5.2.1 `map52` P6/P7/S5/CC/NC/PP paths, not merely to routines present in the directory.

References

[1] Jin, Y., Kewley, L. J., & Sutherland, R. S. 2022, "Messenger Monte Carlo MAPPINGS V (M^3)–A Self-consistent, Three-dimensional Photoionization Code," ApJ, 927, 37, `https://doi.org/10.3847/1538-4357/ac48f3`.

[2] Jin, Y., Kewley, L. J., & Sutherland, R. S. 2022, "Theoretically Modeling Photoionized Regions with Fractal Geometry in Three Dimensions," ApJL, 934, L1, `https://doi.org/10.3847/2041-8213/ac80f3`.

[3] Sutherland, R. S., & Dopita, M. A. 1993, "Cooling Functions for Low-density Astrophysical Plasmas," ApJS, 88, 253.

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[5] Lucy, L. B. 1999, "Computing Radiative Equilibria with Monte Carlo Techniques," A&A, 344, 282.

[6] Verner, D. A., Ferland, G. J., Korista, K. T., & Yakovlev, D. G. 1996, "Atomic Data for Astrophysics. II. New Analytic Fits for Photoionization Cross Sections of Atoms and Ions," ApJ, 465, 487.